

SHUYU WU

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EDUCATION

University of Waterloo

Waterloo, Ontario, CA

Starting 2026.9

Incoming PhD student in Computer Science. Advisor: Freda Shi

University of Michigan - College of Engineering

Ann Arbor, MI, USA

2024.8 – 2025.12

M.S.E degree in Electrical and Computer Engineering with subplan: Signal, Image Processing and Machine Learning.

University of Michigan - College of Engineering

Ann Arbor, MI, USA

2022.8 – 2024.5

B.S.E degree in Computer Science, minor in Mathematics, graduated on May 2024. GPA: 3.91 / 4.00

Relevant A/A+ course work: Data Structure & Algorithm, Linear Algebra, Intro to Machine Learning, Intro to NLP

Shanghai Jiao Tong University (SJTU) - UM-SJTU Joint Institute

Shanghai, China

2020.9 – 2024.8

B.E degree in Electrical and computer engineering (ECE), graduated on August 2024. GPA: 3.63 / 4.00

Relevant A/A+ course work: Programming & Elem. Data Structures, Probabilistic Method in Engineering

PUBLICATION

1. Xiaoxi Luo*, **Shuyu Wu***, Ryker Lin, Yixuan Wang, Ziqiao Ma, Joyce Chai, Freda Shi. Towards the Scaling Law of Neural Model Probes. Under Review
2. **Shuyu Wu***, Ziqiao Ma*, Xiaoxi Luo*, Yidong Huang, Josue Torres-Fonseca, Freda Shi, Joyce Chai. *Tracing the Emergence of Symbol Grounding in Multimodal Language Models*. In proceedings of the Forty-Third International Conference on Machine Learning (ICML 2026)
3. Hala Sheta*, Eric Haoran Huang*, **Shuyu Wu***, Ilia Alenabi, Jiajun Hong, Ryker Lin, Ruoxi Ning, Daniel Wei, Jialin Yang, Jiawei Zhou, Ziqiao Ma, Freda Shi. *From Behavioral Performance to Internal Competence: Interpreting Vision-Language Models with VLM-Lens*. In proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025) System Demonstration

COMMUNITY SERVICE AND CONTRIBUTION

Reviewer

- NeurIPS (CogInterp workshop, 2025)
- ICLR (2026)
- ICML (2026)

Contributor

- Main contributor and code reviewer of the toolkit *VLM-Lens*
- Data contributor of the benchmark *Humanity's Last Exam* (Phan Et al.)

TEACHING EXPERIENCE

University of Michigan (4), Ann Arbor, MI, USA

2023.9 – 2025.12

Grader in ECE 551, Matrix Methods for Signal Processing, Data Analysis and Machine Learning; Grader in CSE 595, Natural Language Processing; Proof tutor in MATH 217, Linear Algebra; Grader in EECS 445, Introduction to Machine Learning.

Shanghai Jiao Tong University (3), Shanghai, China

2021.9 – 2022.12

Teaching assistant in MATH 2860J, Honor Mathematics IV; ECE 4010J, Probabilistic Methods in Engineering; ENGR 1010J, Introduction to Computers and Programming at UM-SJTU joint institute.

RESEARCH EXPERIENCE

SLED (Situated Language and Embodied Dialogue) Lab at University of Michigan 2024.9 –

Research Assistant Supervisor/Mentor: Prof. Joyce Chai, Dr. Ziqiao Ma

- Research interest: Mechanistic Interpretability of language/vision language model

INTERN & PROJECT EXPERIENCE

Capstone Design at Shanghai Jiao Tong University 2024.5 – 2024.8

Supervisor: Kexin Weng from Amore AI

- Collected a synthetic dataset simulating therapeutic dialogues between psychologists and teenagers for fine-tuning LLMs
- Fine-tuned LLMs using LoRA; integrate chain-of-thought (CoT) reasoning to provide personalized mental health support
- Developed audio input/output modules to enable naturalistic voice-based interaction in the prototype system

Course Project (Introduction to Operating Systems) 2024.1 – 2024.5

Use C++ to implement part of an operating system, including

- Thread scheduling, including the implementation of locks and conditional variables
- Virtual memory management, including the implementation of pagers and page replacement mechanisms
- Distributed file systems, including index nodes for files and folders, and system control to solve the problem of multiple users accessing the system at the same time

Data.algebra.ai Inc. 2024.1 – 2024.5

Data Engineer Supervisor: Arif Rahman

Data engineer intern at Data.algebra.ai Inc.

- Collected dataset on credit card transactions
- Design and train a model to identify credit card fraud transaction based on transaction data

Diversity Policy Inc. 2023.5 – 2023.8

Backend and Algorithm Engineer Supervisor: Dr. Brandon Chicotsky, Arif Rahman

Software engineer intern at Diversity Policy Inc.

- Implemented a web crawler to automatically get universities' latest policy and a search engine to search for policy in database
- Developed and maintain a Django server for the official website of the company
- Used AWS services (EC2, S3, etc.) and Docker to deploy the server
- Utilized a ML and NLP approach to classify policy and non-policy webpages

SKILLS

- Programming Languages: C++, Python, Matlab, JavaScript, Julia, Java, swift
- Database: SQLite, MySQL, PostgreSQL, MongoDB
- Libraries and Frameworks: pandas, scikit-learn, tensorflow, pytorch, transformers, django, boto3, whoosh, flask, spring boot
- Mobile Development: Android, ios
- Development Tools: git, GitHub, Docker, nginx
- Web/AI services: Amazon AWS (including EC2, S3 and SNS), OpenAI API

HONORS AND AWARDS

Summa Cum Laude (Graduate with the highest distinction), University of Michigan 2024.5

UM-SJTU Joint Institute Advanced Teaching Assistant Certificate 2023.4

SJTU Undergraduate Excellent Scholarship, 3rd prize 2021.11

LINKS

- GitHub: <https://github.com/wonderwind271>
- Huggingface: <https://huggingface.co/wonderwind271>